

Digital Gold Wallet Application

Application API Endpoints

User & Address Management APIs

Endpoints

- **POST** /api/v1/users – Create new user
 - **GET** /api/v1/users/{userId} – Get user details
 - **GET** /api/v1/users – List users (pagination supported)
 - **PUT** /api/v1/users/{userId} – Update user profile
 - **GET** /api/v1/users/{userId}/balance – Get wallet balance
 - **POST** /api/v1/addresses – Create address
 - **GET** /api/v1/addresses/{addressId} – Get address details
 - **PUT** /api/v1/addresses/{addressId} – Update address
-

Vendor & Branch Management APIs

Endpoints

- **POST** /api/v1/vendors – Create gold vendor
 - **GET** /api/v1/vendors/{vendorId} – Get vendor details
 - **GET** /api/v1/vendors – List all vendors
 - **PUT** /api/v1/vendors/{vendorId} – Update vendor profile
 - **GET** /api/v1/vendors/{vendorId}/price – Get current gold price
 - **POST** /api/v1/vendors/{vendorId}/branches – Add vendor branch
 - **GET** /api/v1/branches/{branchId} – Get branch details
 - **GET** /api/v1/vendors/{vendorId}/branches – List vendor branches
 - **GET** /api/v1/branches/{branchId}/inventory – Get gold quantity at branch
-

Wallet, Payments & Balance APIs

Endpoints

- **POST** /api/v1/wallets/{userId}/credit – Add money to wallet
 - **POST** /api/v1/wallets/{userId}/debit – Debit money from wallet
 - **GET** /api/v1/wallets/{userId}/balance – Wallet balance
 - **POST** /api/v1/payments – Initiate payment
 - **GET** /api/v1/payments/{paymentId} – Payment status
 - **GET** /api/v1/users/{userId}/payments – Payment history
-

Virtual Gold Holdings APIs

Endpoints

- **POST** /api/v1/gold/virtual/buy – Buy virtual gold
 - **POST** /api/v1/gold/virtual/sell – Sell virtual gold
 - **GET** /api/v1/users/{userId}/gold/virtual – User's virtual gold holdings
 - **GET** /api/v1/branches/{branchId}/gold/virtual – Branch-wise holdings
-

Physical Gold & Transaction History APIs

Endpoints

- **POST** /api/v1/gold/physical/convert – Convert virtual to physical gold
 - **GET** /api/v1/users/{userId}/gold/physical – Physical gold orders
 - **GET** /api/v1/physical-transactions/{transactionId} – Delivery details
 - **GET** /api/v1/users/{userId}/transactions – Complete transaction history
 - **GET** /api/v1/branches/{branchId}/transactions – Branch transaction log
-

High-Impact Reporting Endpoints

✓ 1. Vendor Performance Report

Endpoint

GET /api/v1/reports/vendors/performance

Digital Gold Wallet Application

Business Purpose

- Measures which vendors are generating revenue
- Helps in business decisions (partnerships, scaling, pricing)

Key Metrics

- Total transactions
- Total gold volume traded
- Total revenue generated

Sample Response

JSON

```
[
{
"vendorId": 1,
"vendorName": "Sona Jewelers",
"totalTransactions": 120,
"totalVolume": 2500.75,
"totalRevenue": 16000000
}
]
```

Tables Used

- vendors
- vendor_branches
- transaction_history

Why This is Strong

- ✓ Requires multi-table joins
- ✓ Aggregation (SUM, COUNT)
- ✓ Real business KPI understanding

✓ 2. Branch Inventory vs User Holdings

Endpoint

GET /api/v1/reports/branches/{branchId}/inventory-status

Business Purpose

- Validates gold availability vs allocated gold
- Prevents overselling / inventory mismatch

Key Metrics

- Total available gold (branch inventory)
- Allocated gold (sum of user holdings)
- Remaining stock

Sample Response

JSON

```
{
"branchId": 5,
"availableGold": 1200,
"allocatedToUsers": 850,
"remainingGold": 350
}
```

Tables Used

- vendor_branches (inventory)
- virtual_gold_holdings (user allocation)

Why:

- ✓ Tests business logic (not just queries)
- ✓ Introduces real-world reconciliation problem
- ✓ Evaluates understanding of data consistency

Digital Gold Wallet Application

✓ 3. Top Investors Report

Endpoint

GET /api/v1/reports/users/top-investors?limit=10

Business Purpose

- Identifies high-value customers (HNI segment)
- Useful for marketing, rewards, and retention strategies

Key Metrics

- Total investment per user (based on transactions)
- Ranking of users

Sample Response

JSON

```
[
{
  "userId": 21,
  "name": "Pradeep Kumar",
  "totalInvestment": 200000
}
```

Tables Used

- users
- transaction_history

Why:

- ✓ Sorting + aggregation
- ✓ Pagination (limit)
- ✓ Business insight (customer segmentation)